

Assignment No. 1: Shopping Basket Problem

The sales report of a store's products over a period of time is provided in the file `factors.csv`, containing the following information:

- Order_id: Invoice number
- Product_id: Product identifier
- Product_name: Product name
- Department_id: Department identifier
- Department_name: Department name

A)

As a data analyst, you are required to provide two reports as follows:

1. Which products do store customers commonly buy together?
2. Which departments' products do store customers commonly purchase together in their shopping baskets?

Your report should include the names of product pairs (and department pairs for the second report), along with their **confidence** and **support** values.

B)

Later, the store decided to manage each department independently. You are asked to provide the following report for each department:

- Which products do store customers commonly buy together?

For all the above cases, a minimum support of 0.003 (0.3%) and a minimum confidence of 0.25 (25%) should be considered.

A sample of Report No. 1 for the first 50% of the data is provided in the file `report.csv`. Note that this file only includes product codes, while the managers also need the product names.